GENERAL NOTES: UNLESS OTHERWISE SPECIFIED

1. GENERAL & STANDARDS COMPLIANCE:
1.1 INTERPRET DRAWING IN ACCORDANCE WITH ASME Y14.5-2018 AND ISO 8015.
1.2 3D CAD DATASETS GOVERN NOMINAL GEOMETRY:
- NEAR-NET SLM ADDITIVE BLANK: W3-AERO-THERMAL-CORE-8.2_NEAR_NET_SLM.STEP
- FINAL FINISHED CNC COMPONENT: W3-AERO-THERMAL-CORE-8.2_FINAL_CNC.STEP
1.3 GENERAL TOLERANCES: LINEAR ISO 2768-m (±0.30 mm), ANGULAR ±0.1 DEG.
1.4 THIRD ANGLE PROJECTION APPLIES THROUGHOUT. ALL UNITS IN MILLIMETERS.
1.5 COORDINATE REFERENCE FRAME: FIRST OCTANT CONVENTION (ALL POSITIVE X, Y, Z FROM DATUM C-D-A).

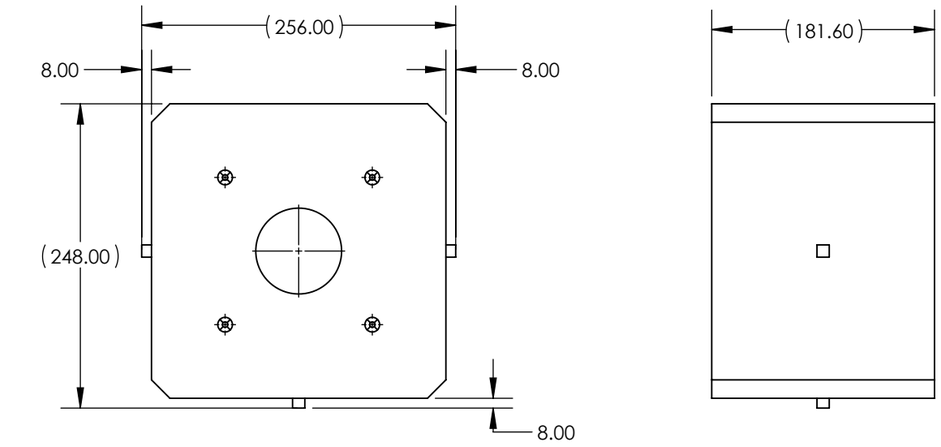
2. ADVANCED METALLURGY & ADDITIVE (SLM) PROTOCOL:
2.1 MATERIAL: PRECIPITATION-HARDENED CHROMIUM-ZIRCONIUM COPPER ALLOY (CuCr1Zr).
2.2 ADDITIVE PROCESS: GREEN LASER PBF-LB/M (515 nm WAVELENGTH, MINIMUM OPTICAL ABSORPTIVITY >= 70%) TO ELIMINATE REFLECTIVITY SPATTER AND PREVENT KEYHOLE VOID POROSITY.
2.3 CONFORMAL FLUID CHANNEL GEOMETRY: ALL HORIZONTAL PASSAGES FEATURE INTEGRAL SELF-SUPPORTING TEARDROP ARCHES (45-DEGREE ANGLE TO BUILD AXIS +Z). ZERO INTERNAL SUPPORTS PERMITTED.
2.4 DEPOWDERING MANDATE: UNMELTED FEEDSTOCK POWDER MUST BE 100% EVACUATED THROUGH 4X Ø4.50 mm BOTTOM PORTS VIA PNEUMATIC FLUIDIZATION AND VIBRATORY EXCITATION BEFORE HEAT TREATMENT.

3. POST-PROCESSING, HIP & THERMAL CONDITIONING:
3.1 STRESS RELIEF: HEAT TREAT ON ORIGINAL BUILD PLATE IN INERT ARGON ATMOSPHERE AT 500-550 DEG C PRIOR TO WIRE-EDM CUTOFF TO RELAX RESIDUAL TENSILE STRESSES.
3.2 HOT ISOSTATIC PRESSING (HIP): MANDATORY AT 900 DEG C, 100-150 MPa ARGON PRESSURE TO COLLAPSE RESIDUAL MICRO-VOIDS, YIELDING DENSITY > 99.95% AND THERMAL CONDUCTIVITY >= 305 W/m·K.
3.3 INTERNAL CHANNEL SMOOTHING: ABRASIVE FLOW MACHINING (AFM) APPLIED THROUGH FLUID PORTS TO REDUCE INTERNAL WALL ROUGHNESS FROM AS-BUILT Ra 10-15 µm DOWN TO Ra <= 1.0 µm.

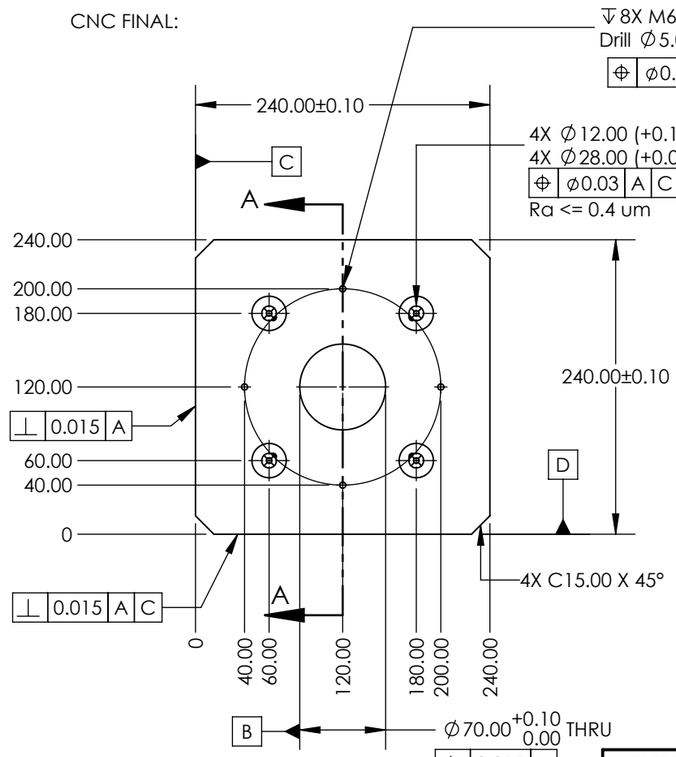
4. 5-AXIS CNC FINISHING & GD&T CONTROLS:
4.1 DATUM ALIGNMENT: USE MACHINE TOUCH PROBE (RENISHAW) TO LOCATE 3X SACRIFICIAL DATUM PADS ON NEAR-NET PREFORM TO ESTABLISH WORK COORDINATE SYSTEM G54 PRIOR TO FINISH MILLING.
4.2 PRIMARY DATUM [A] (BOTTOM BASE): FLATNESS <= 0.010 mm ACROSS 240x240 mm ENVELOPE. SURFACE FINISH Ra <= 0.8 MICRONS.
4.3 SECONDARY DATUM [B] (CENTER BORE Ø70.00 THRU): PERPENDICULAR TO [A] WITHIN 0.015 mm.
4.4 4X C-RING SEAL COUNTERBORES: DIAMETER Ø28.00 +0.08/0.00 mm, DEPTH 3.20 ±0.05 mm. TRUE POSITION WITHIN Ø0.03 mm RELATIVE TO [A], [C], [D]. SEALING LAND Ra <= 0.4 MICRONS (HELIUM LEAK RATE < 1.0E-9 mbar·L/s).
4.5 8X PCD M6 TAPPED HOLES: CENTERED ON PCD Ø160.00 mm, TRUE POSITION WITHIN Ø0.05 mm RELATIVE TO [B], [C], [D].

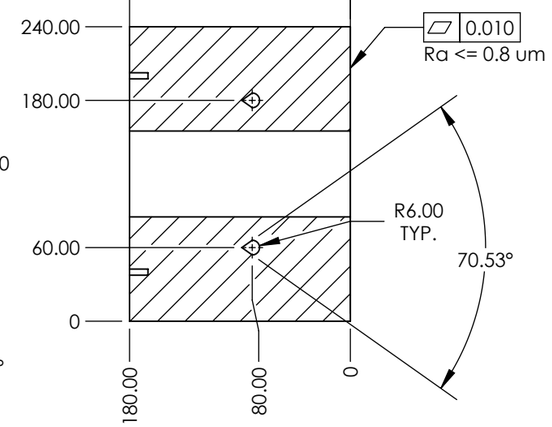
5. QUALITY VERIFICATION & PRESSURE PROOF:
5.1 HYDROSTATIC PROOF PRESSURE: 100% PRESSURE PROOF TEST AT 25.0 MPa (3625 PSI) WITH WATER/GLYCOL FOR 10 MINUTES. ZERO LEAKAGE OR PERMANENT DEFORMATION ALLOWED.
5.2 CLEANLINESS: PASSIVATE AND RINSE WITH DEIONIZED ULTRAPURE WATER, SEAL IN VACUUM-PURGED PACKAGING FOR AEROSPACE PROPULSION DEPLOYMENT.

NEAR NET SLM:

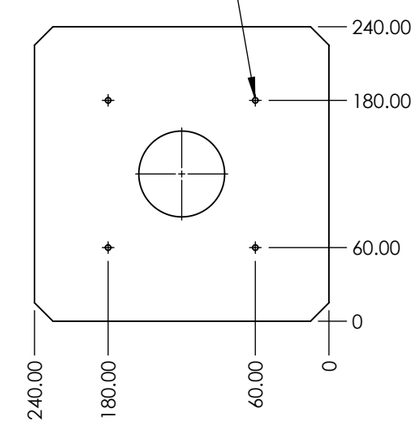


CNC FINAL:





SECTION A-A



UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: Linear ISO 2768-m (±0.30 mm), Angular ±0.1°	FINISH: Abrasive Flow Machining (AFM) for internal channels (Ra <= 1.0 µm); Exterior raw machined / passivated		DEBURR AND BREAK SHARP EDGES		DO NOT SCALE DRAWING	REVISION: A
					Mode 5: Hybrid SLM (Green Laser 515nm) + HIP + 5-Axis CNC Finishing	
	NAME	SIGNATURE	DATE		TITLE: HYPERSONIC CONFORMAL-COOLED COPPER MANIFOLD	
DRAWN						
CHK'D						
APPV'D						
MFG						
Q.A				MATERIAL: Precipitation-Hardened Chromium-Zirconium Copper Alloy CuCr1Zr (AMS 4504 equivalent)	DWG NO. W3-AERO-THERMAL-CORE-8.2	A3
				WEIGHT: ~14.18 kg	SCALE NTS	SHEET 1 OF 1